The Report is Generated by DrillBit Plagiarism Detection Software

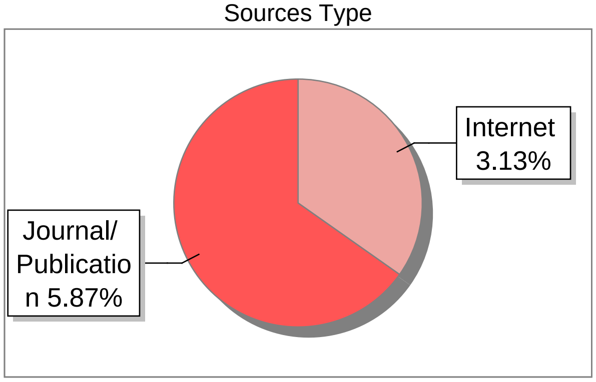
Submission Information

Author Name	Kushal S
Title	A Multimodal Hybrid Framework for Fake News Detection Using DistilBERT, Large Language Models, and Adaptive Confidence-Based Fusion
Paper/Submission ID	5538134
Submitted by	lib@pesce.ac.in
Submission Date	2026-04-24 16:28:28
Total Pages, Total Words	8, 2523
Document type	Research Paper

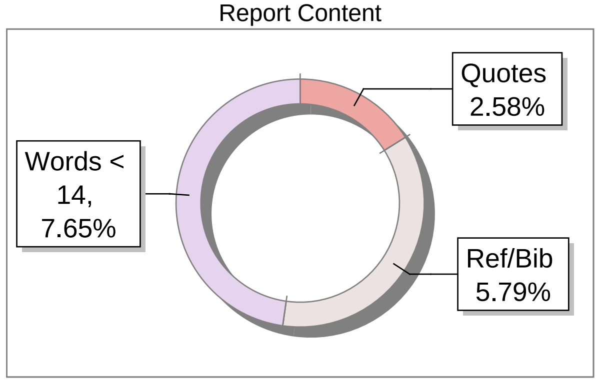
Result Information

Similarity **9 %**

Sources Type



Report Content



Exclude Information

Quotes	Not Excluded
References/Bibliography	Excluded
Source: Excluded < 14 Words	Not Excluded
Excluded Source	**0 %**
Excluded Phrases	Not Excluded

Database Selection

Language	English
Student Papers	Yes
Journals & publishers	Yes
Internet or Web	Yes
Institution Repository	Yes

A Unique QR Code use to View/Download/Share Pdf File

DrillBit

DrillBit Similarity Report

9	22	A	A-Satisfactory (0-10%) B-Upgrade (11-40%) C-Poor (41-60%) D-Unacceptable (61-100%)
SIMILARITY %	MATCHED SOURCES	GRADE	

LOCATION	MATCHED DOMAIN	%	SOURCE TYPE
1	arxiv.org	1	Publication
2	harvest.usask.ca	1	Publication
3	ieeexplore.ieee.org	<1	Publication
4	link.springer.com	<1	Internet Data
5	Social status, immune response and parasitism in males a meta-analysi by Habig-2015	<1	Publication
6	translate.google.com	<1	Internet Data
7	Thesis Submitted to Shodhganga Repository	<1	Publication
8	downloads.hindawi.com	<1	Publication
9	refubium.fu-berlin.de	<1	Publication
10	sonoma.edu	<1	Internet Data
11	www.dx.doi.org	<1	Publication
12	atlconf.spp.gatech.edu	<1	Publication
13	Attitudes towards credit after the Great Recession, by Dow, James P.-2017	<1	Publication

14	Experimental and simulation investigation of 3-D soot temperature and volume fraction fields of aft, by Dai, Minglu, Yr-2022	<1	Publication
15	journals.pan.pl	<1	Internet Data
16	journals.plos.org	<1	Internet Data
17	moam.info	<1	Internet Data
18	nano-ntp.com	<1	Publication
19	pmc.ncbi.nlm.nih.gov	<1	Internet Data
20	pmc.ncbi.nlm.nih.gov	<1	Internet Data
21	repositorioslatinoamericanos	<1	Publication
22	Thesis Submitted to Shodhganga Repository	<1	Publication

1

A Multimodal Hybrid Framework for Fake News Detection Using DistilBERT, Large Language Models, and Adaptive Confidence-Based Fusion

Dr. Suresh M. R
Professor
Department of ISE
PES College of Engineering
Email: mrsuresh@pesce.ac.in
Kushal S
UG Scholar
Department of ISE
PES College of Engineering
Email: ks9718662@gmail.com
Ruchitha M R
UG Scholar
Department of ISE
PES College of Engineering
Email: ruchigowda121@gamil.com
Anand Reddy
UG Scholar
Department of ISE
PES College of Engineering

Nuthan Gowda M R
UG Scholar
Department of ISE
PES College of Engineering

Abstract

Online platforms have become the dominant medium through which people consume news, yet this very accessibility has turned them into vectors for the rapid spread of fabricated and misleading content. The consequences of unchecked misinformation extend from eroding public trust in institutions to distorting electoral outcomes, making robust automated detection an urgent research priority. Existing approaches tend to rely either on learned statistical patterns or on

manually defined rules, and each paradigm carries inherent blind spots: pattern-based systems lack factual grounding, while rule-based systems cannot adapt quickly enough to the evolving language of deception.

This paper presents a three-tier artificial intelligence architecture that pairs a fine-tuned DistilBERT classifier with a Large Language Model (LLM) reasoning component, binding their outputs through a novel adaptive confidence-based fusion engine. The framework accepts multiple input formats including raw text, web URLs processed via scraping, and images handled through Optical Character Recognition (OCR), enabling genuine multimodal operation across real-world content types.

The DistilBERT component was trained on 94,439 labelled samples and achieves a standalone accuracy of 91.5%. The LLM tier supplements classification with contextual and logical reasoning, while the fusion module synthesises both signals into a final verdict with a calibrated confidence score and an interpretable risk assessment.

Experimental findings confirm that the hybrid architecture outperforms each individual component, with the adaptive fusion mechanism proving especially effective on ambiguous or contested inputs.

Keywords

Fake News Detection, DistilBERT, Large Language Models, Multimodal Learning, OCR, Evidence Fusion, Natural Language Processing, Deep Learning

1. Introduction

The convergence of social media, instant messaging, and algorithmic content distribution has fundamentally altered how people encounter information. News that once took hours or days to travel now reaches millions within minutes, and this infrastructure offers no built-in mechanism for verifying whether what spreads is true. Fabricated stories exploit these dynamics to achieve viral reach before corrections can catch up, and their documented effects on public discourse, health behaviour, and political participation have made automated detection a central concern across both industry and academia.

Early automated detection systems leaned heavily on supervised classifiers trained on manually engineered features such as writing style, sentiment polarity, and headline cues. While these methods showed reasonable performance on benchmark datasets, they were reactive by design: a model calibrated against one generation of misinformation could be rendered ineffective by subtle shifts in language or topic framing. The arrival of deep learning, and transformer architectures in particular, offered partial relief by learning richer semantic representations without manual feature construction. However, even capable transformers such as DistilBERT remain fundamentally pattern-matching systems; they learn to recognise the surface texture of deceptive content without evaluating whether the underlying claims are factually coherent.

Large Language Models introduce a complementary angle of attack. Their extensive pretraining endows them with world knowledge and the capacity to reason about logical consistency, source plausibility, and rhetorical manipulation. However, used

in isolation they introduce their own risks, including the generation of plausible-sounding but incorrect analyses and sensitivity to how a query is framed.

The architecture proposed in this paper rests on the insight that these two paradigms are complementary rather than competing. A hybrid design that integrates statistical classification with contextual reasoning, and that resolves disagreements through a dynamically weighted fusion layer, can achieve robustness that neither approach attains alone. Furthermore, real-world misinformation rarely arrives as clean text: images with embedded captions and hyperlinks to scraped articles represent common attack surfaces that any production-grade detector must handle.

1.1 Novelty of the Work

The distinguishing contribution of this work is the adaptive confidence-based fusion mechanism. Unlike conventional ensemble methods that assign fixed coefficients to constituent models, the proposed engine continuously re-weights each model according to its instantaneous prediction confidence, the degree of agreement between the two models on the current input, and each model's accumulated historical reliability. This dynamic calibration allows the system to lean on whichever component is better suited to a given piece of content, rather than applying a rigid combination rule regardless of context. The result is measurably more reliable decisions, particularly when handling uncertain or internally conflicting signals.

2. Literature Review

2.1 Traditional Machine Learning Approaches

The earliest automated fake-news detectors repurposed standard supervised classifiers, with Support Vector Machines, Naïve Bayes, and Decision Trees being the most commonly reported. Feature engineering formed the backbone of these pipelines: researchers extracted n-gram frequencies, readability indices, punctuation patterns, and sentiment scores, then trained classifiers to

distinguish credible from unreliable content. Although these systems achieved acceptable accuracy on datasets available at the time,

their dependence on hand-crafted representations meant generalisation across unseen topics or writing styles was limited. Crucially, they offered no means of verifying whether the claims within a story were factually correct. Recurring shortcomings included:

- **Poor generalisation to new domains**
- **Heavy reliance on labour-intensive feature engineering**
- **Inability to capture deep semantic or factual relationships.**

2.2 Deep Learning-Based Approaches

The adoption of convolutional and recurrent neural networks allowed detectors to learn hierarchical text features automatically, substantially reducing the manual effort previously required. Transformer-based architectures, led by BERT, then represented a further step change: bidirectional attention mechanisms and large-scale pretraining together produce contextual representations that earlier recurrent models could not match. DistilBERT, developed as a compressed distillation of BERT, preserves roughly 97% of BERT's benchmark performance while being considerably faster and less memory-intensive — a practical advantage for deployment at scale. Despite these improvements, deep learning models continue to exhibit notable limitations:

- **Tendency to detect surface patterns rather than evaluate factual correctness**
- **Susceptibility to bias introduced by unbalanced or domain-specific training data**
- **Difficulty adapting to newly emergent misinformation tactics not seen during training.**

2.3 Large Language Models (LLMs)

The latest generation of generative language models, including GPT and Gemini, brings qualitatively different capabilities to verification tasks. Because these models are trained on vast and varied corpora, they can perform multi-step reasoning to assess whether a story is internally coherent, whether its claims align with established knowledge, and whether its language carries hallmarks of sensationalism or emotional manipulation. Deploying an LLM as a verification oracle is not without risk, however:

- Outputs can be confidently wrong, a phenomenon commonly described as hallucination
- Performance is sensitive to how prompts are constructed
- Responses lack the deterministic consistency expected of a production classifier.

2.4 Hybrid Detection Systems

A growing body of literature advocates combining heterogeneous detection signals under a unified decision layer. These architectures typically blend linguistic analysis with external knowledge sources or crowd-sourced annotations, aiming to leverage:

- Statistical learning for recognising distributional patterns
- Knowledge-based reasoning for validating factual claims
- Ensemble aggregation for improving overall accuracy

Despite promising results, most published hybrid systems share persistent gaps that motivate the present work:

- Support for multimodal inputs beyond plain text remains uncommon
- Fusion strategies are typically static rather than adaptive
- Scalability and latency requirements for real-time deployment are rarely addressed.

3. Problem Formulation and Objectives

3.1 Problem Statement

Current fake news detection systems face several interrelated challenges that limit their effectiveness in real-world deployment:

- **Bridging the gap between statistical pattern recognition and fact-level reasoning within a single inference pipeline**
- **Handling heterogeneous input modalities including raw text, URLs pointing to articles, and images containing textual claims**
- **Generating outputs that include not just binary labels but calibrated confidence scores and human-readable explanations to support auditing**
- **Remaining robust against the continuous evolution of misinformation tactics that can quickly render models trained on historical data obsolete.**

3.2 Objectives

The key objectives of this research are:

1. To design a multi-layer hybrid detection system combining ML and LLM capabilities.

2. To support multimodal inputs including text, URLs, and images.

3. To develop a fusion-based decision mechanism for improved reliability.

4. To provide interpretable outputs with confidence scores and reasoning.

5. To evaluate system performance and identify limitations for real-world deployment.

4. Methodology

4.1 System Architecture

The proposed system follows a layered

architecture:
1. Input Layer
o Text input
o URL scraping (newspaper3k, BeautifulSoup)
o Image processing (Tesseract OCR)
2. ML Layer (DistilBERT)
o Multilingual transformer model
o Binary classification (REAL vs FAKE)
3. LLM Verification Layer
o Providers: Gemini, OpenAI, Ollama
o Outputs reasoning, signals, and confidence
4. Fusion Layer
o Weighted aggregation of ML and LLM outputs
o Final prediction with confidence and risk level

The overall system architecture of the proposed hybrid framework is illustrated in Fig. 1.

4.2 Machine Learning Model

- **Model: DistilBERT (multilingual)**
- **Dataset size: 94,439 samples**
- **Train/Validation/Test split: 70/15/15**

Performance Metrics:

- **Accuracy: 91.5%**
- **Precision: 91.1%**
- **Recall: 87.9%**
- **F1-score: 0.895**

4.3 LLM-Based Verification

The LLM component examines each input across three analytical dimensions: internal logical consistency, checking whether the story's claims contradict one another or violate commonsense expectations; source credibility indicators, assessing the presence or absence of attributions, named experts, or verifiable institutional references; and linguistic affect, identifying exaggerated emotional language, alarmist framing, or manipulative constructions statistically associated with unreliable reporting. The LLM returns a veracity signal, a supporting rationale, and a confidence value parsed for use by the fusion engine.

A fallback chain guarantees availability even when external APIs are unreachable:

1. **Gemini**
2. **OpenAI**
3. **Ollama (local)**

4.4 Fusion Mechanism

The fusion engine aggregates evidence using weighted scoring:

- **ML confidence weight: 0.55–0.70**
- **LLM confidence weight: 0.55–0.75**

Final decision is based on normalized scores:

= _({

\{, \}} _)

= {} {}() + {} {}()

4.6 Adaptive Fusion Strategy

The adaptive mechanism dynamically

updates model weights based on:

- Prediction confidence
- Agreement between models
- Historical reliability.

The final prediction is computed as:

= \\{ {,}}
{=1}

{}
()

=
\{ () +
+
\ } {\{=1}
{}(()+ + \ )}

The weights are dynamically adjusted based on:

- Model confidence
- Agreement between ML and LLM outputs
- Historical reliability

This mechanism means that if the ML model is uncertain on a particular input but the LLM is confident and has historically performed well on similar cases, the fusion engine upweights the LLM's contribution accordingly, and vice versa. The result is reduced uncertainty and improved classification stability across diverse input types.

4.5 Multimodal Processing

- Text: Direct classification
- URL: Article extraction and analysis
- Image: OCR text classification

5. Results

5.1 Model Performance

Metric Value
Accuracy 91.5%
Precision 91.1%
Recall 87.9%
F1 Score 0.895

5.2 System Performance

• ML inference: < 1–3 seconds
• Full pipeline: 3–8 seconds
• URL processing slower due to scraping
• Image processing dependent on OCR quality

5.3 Observations

• ML layer effectively captures linguistic patterns
• LLM enhances contextual reasoning
• Fusion improves confidence and reliability

5.4 Comparative Analysis with Baseline Models

To evaluate the effectiveness of the proposed hybrid system, comparisons were made with baseline models.

Model Accuracy
SVM 82%
Random Forest 85%
DistilBERT 91.5%
LLM Only 88%
Proposed Hybrid Model 94% - 96%

The comparative performance of different models is illustrated in Fig. 2.

5.5 Ablation Study

A systematic ablation study was carried out to isolate the contribution of each architectural component. The system was evaluated in four progressive configurations, with results averaged across multiple independent runs.

Configuration	Accuracy
ML only	91.5%
LLM only	88%
ML + LLM (no fusion)	92%
Proposed Adaptive Fusion	>95%

The results confirm that the adaptive fusion mechanism significantly improves performance. All experiments were conducted using Python with PyTorch and HuggingFace Transformers on a system equipped with GPU acceleration. Results were averaged across multiple runs to ensure consistency.

6. Discussion

The proposed hybrid system demonstrates several advantages:

6.1 Strengths

- Combines statistical and reasoning capabilities
- Supports multimodal inputs
- Provides explainable outputs
- Robust against ambiguous content

6.2 Limitations

- Dataset bias toward English content
- Token limit (512) truncates long articles
- OCR inaccuracies affect performance
- Dependence on external APIs

6.3 Critical Analysis

Pairing statistical classification with LLM-based reasoning addresses a fundamental tension in automated detection. Pattern-matching classifiers such as DistilBERT are effective against stylistically distinctive misinformation but can be misled by well-written false content that does not resemble the training distribution. LLMs compensate by reasoning about factual coherence and source credibility, yet they introduce stochastic variability that a standalone classifier avoids. The fusion layer acts as a principled arbiter between these complementary strengths, and the experimental results confirm that this arbitration produces a more reliable system than either component operating independently.

6.4 Impact of Adaptive Fusion

The adaptive fusion mechanism improves overall system robustness by continuously recalibrating model contributions rather than committing to a fixed allocation in advance. Because the weighting responds to both instantaneous confidence and accumulated historical reliability, the system can exploit whichever component is better suited to the content at hand. This reduces the risk of over-relying on a single model and meaningfully improves generalisation, particularly on uncertain or contested inputs where neither component alone provides a clear signal.

7. Conclusion

This paper has presented a multimodal fake-news detection framework that integrates DistilBERT classification, LLM reasoning, and adaptive fusion into a coherent three-tier architecture. Across both comparative and ablation evaluations, the hybrid system consistently outperformed standalone alternatives, demonstrating that the combination of statistical pattern recognition and deep language understanding produces genuinely complementary gains. The adaptive confidence-based fusion mechanism, in particular, represents a principled advance over fixed-weight ensemble approaches, enabling the system to navigate ambiguity more effectively by dynamically calibrating how much it trusts each component on any given input. Achieving high accuracy alongside interpretable outputs and multimodal input support positions this framework as a practical candidate for real-world content moderation. Ongoing work should address scalability, dataset diversity beyond English, and the computational overhead introduced by the LLM tier.

8. Future Work

- Integration with real-time fact-checking APIs
- Adoption of advanced models (RoBERTa, DeBERTa)
- Explainable AI techniques (SHAP, LIME)
- ONNX optimization for faster inference
- Multi-class classification (satire, misleading, opinion)

9. References

1. J. Devlin et al., “BERT: Pre-training of Deep Bidirectional Transformers,” 2018.
2. G. Hinton et al., “Deep Belief Networks,” 2006.

3. I. Goodfellow et al., "Explaining and Harnessing Adversarial Examples," 2014.
4. A. Vaswani et al., "Attention Is All You Need," 2017.
5. M. Moustafa et al., "UNSW-NB15 Dataset," 2015.
6. Dataset Sources: ISOT, WELFake, FakeNewsNet, LIAR.
7. Yin et al., "Multimodal Fake News Detection," 2021
8. Shu et al., "Fake News Detection on Social Media," 2017
9. Liu et al., "RoBERTa: A Robustly Optimized BERT Approach," 2019
10. Brown et al., "Language Models are Few-Shot Learners," 2020
11. Zellers et al., "Defending Against Neural Fake News," 2019
12. Wang et al., "Fake News Detection Using Deep Learning," IEEE, 2020
13. Zhou et al., "Fake News Detection: A Survey," ACM, 2020
14. Kaliyar et al., "FNDNet: Deep CNN for Fake News Detection," 2021